

ERCOT Basis Risk & DA/RT Convergence Analysis

Analyst Research Memo | Garrison Cockrell | Texas A&M University
Sample Period: January 1, 2024 – June 6, 2026 | ERCOT HB_NORTH & HB_WEST

OBJECTIVE

This memo analyzes two related sources of price risk in the ERCOT wholesale power market: (1) day-ahead to real-time (DA/RT) convergence risk at HB_NORTH and HB_WEST, and (2) North-West hub basis risk across both the day-ahead market (DAM) and real-time market (RTM). The analysis incorporates physical system driver attribution — linking extreme price outcomes to net load, renewable output, and net-load ramping. Findings are framed as market-risk attribution, not a trading strategy.

DA/RT CONVERGENCE RISK — PHYSICAL DRIVER ATTRIBUTION

Both hubs showed modest positive average DA/RT spreads (\$1.04/MWh at HB_NORTH, \$1.17/MWh at HB_WEST) across the full sample. The central risk is not the average — DA/RT convergence exposure is dominated by low-frequency, high-severity RTM price spikes. The worst 1% of outcomes averaged -\$214.49/MWh at HB_NORTH and -\$222.07/MWh at HB_WEST, occurring during materially elevated net-load and suppressed renewable-share conditions.

Hub	Condition	Avg DA/RT Spread (\$/MWh)	Avg Net Load (MW)	Renew. Share	Avg Load Ramp (MW)
HB_NORTH	Full Sample	\$1.04	35,163	36%	0.47
HB_NORTH	Worst 1% DA/RT	-\$214.49	48,088	18%	1,445
HB_WEST	Full Sample	\$1.17	35,163	36%	0.47
HB_WEST	Worst 1% DA/RT	-\$222.07	47,368	16%	1,381

- Worst 1% net load at HB_NORTH averaged 48,088 MW — 37% above the full-sample mean of 35,163 MW.
- Renewable share collapsed from 36% (full sample) to 18% at HB_NORTH and 16% at HB_WEST during worst 1% outcomes.
- Average net-load ramp during worst 1% outcomes was 1,445 MW/hr (HB_NORTH) and 1,381 MW/hr (HB_WEST), versus 0.47 MW/hr full-sample — confirming rapid thermal demand acceleration as a stress indicator.
- Worst single event: August 20, 2024 Hour 20 — DA/RT spread of -\$2,400/MWh at HB_NORTH; ERCOT load 81,519 MW, renewable share 14%.

NORTH-WEST HUB BASIS RISK — PHYSICAL DRIVER ATTRIBUTION

The full-sample average RTM North-West basis was -\$1.19/MWh (DAM: -\$1.32/MWh) — modest in average terms. However, RTM basis is materially more volatile than DAM basis, and tail events reflect distinct opposing physical regimes.

Condition	Avg RTM Basis (\$/MWh)	Avg DAM Basis (\$/MWh)	Avg Wind (MW)	Renew. Share	Avg Net Load (MW)
Full Sample	-\$1.19	-\$1.32	12,920	36%	35,163
Top 1% Positive RTM Basis	+\$54.76	+\$18.12	21,670	49%	31,020
Top 1% Negative RTM Basis	-\$71.21	-\$26.22	8,740	20%	39,768
Top 1% Absolute RTM Basis	-\$31.22	-\$13.99	11,932	25%	40,173

- Positive RTM basis events (North above West) occurred during high-wind, high-renewable conditions — avg wind 21,670 MW, renewable share 49%, net load 31,020 MW. Consistent with solar/wind-heavy mid-day dispatch creating regional divergence.

- Negative RTM basis events (West above North) occurred during low-renewable, high net-load conditions — renewable share 20%, net load 39,768 MW. Consistent with thermal dispatch pressure in the West zone.
- RTM basis tail events are 3–4x larger in magnitude than corresponding DAM basis events, confirming that real-time market conditions amplify regional price divergence beyond what the day-ahead market prices in.
- Largest single basis event: March 7, 2024 Hour 21 — RTM North-West basis of -\$260.33/MWh vs. DAM basis of -\$31.44/MWh.

CONCLUSIONS

- DA/RT convergence exposure is a tail-risk problem. Average spreads are modestly positive (\$1.04–\$1.17/MWh), but the worst 1% of outcomes averaged -\$214 to -\$222/MWh — driven by RTM spike events in high net-load, low-renewable regimes.
- RTM North-West basis is a distinct and more volatile source of regional price risk than DAM basis. Tail events are 3–4x larger in RTM than DAM, with opposing physical regimes driving positive versus negative basis outcomes.
- Physical driver attribution provides a risk framework: elevated net load, suppressed renewable share, and positive net-load ramp are the most reliable leading indicators of adverse DA/RT outcomes.
- Both exposures are risk-adjusted spread problems — not arbitrage opportunities — and require explicit tail-risk framing rather than average-spread analysis.

ASSUMPTIONS & LIMITATIONS

- RTM interval prices averaged to hourly frequency to align with DAM settlement data; intra-hour volatility may be smoothed.
- Physical attribution covers January 2024 – December 2025 due to complete annual wind/solar file availability.
- Transaction costs, collateral, bid/offer spreads, and virtual bidding execution constraints are outside scope.
- Historical relationships represent risk attribution, not causal proof or guaranteed forward relationships.

Data: ERCOT Market Information System (DAM/RTM settlement point prices, generation by fuel type, system load). Tools: Python — pandas, NumPy, matplotlib, scipy, statsmodels.